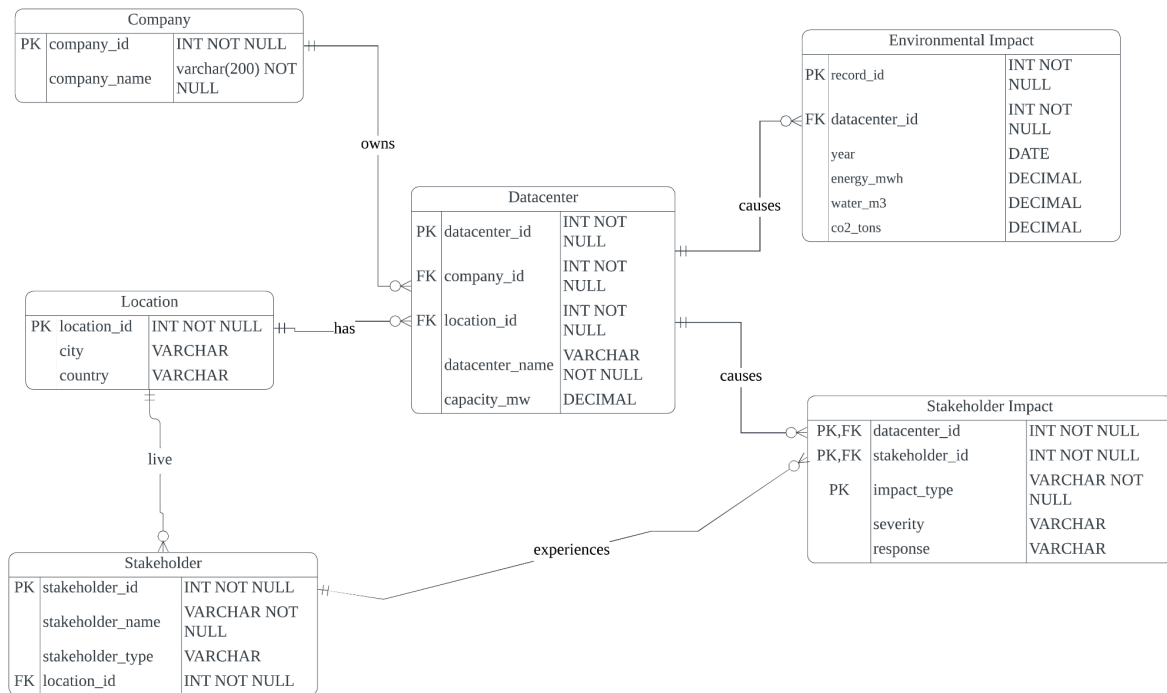


Final ERD (Lucid Chart)



Assignment - Normalization Processes

How our tables evolved.. Violation highlighted in Red, and fixes in Green

Version 1 - No Normalization

DataCenter

dc_id	dc_name	city	country	company_name	year_2023_energy_mwh	year_2023_water_m3	year_2024_energy_mwh	year_2024_water_m3	stakeholder_names	impact_types
1	AMS-DC1	Amsterdam	NL	Google	500000	120000	520000	125000	Residents, Municipality	financial, noise, regulatory
2	AMS-DC2	Amsterdam	NL	Microsoft	300000	80000	310000	82000	Residents	noise

3	RTD-D C1	Rotterd am	NL	Google	400000	100000	420000	105000	NGO, Utility	environ mental, financial
---	-------------	---------------	----	--------	--------	--------	--------	--------	-----------------	---------------------------------

Version 2 - 1NF Fix

What did we fix for 1NF?

- stakeholder_names and impact_types stored multiple values in one cell
- the energy and water data was split into separate columns per year

How we fixed?

- Gave each combination of data center, year, and stakeholder its own row, and remove all list columns

DataCenter

dc_id	year	stake hold er_id	dc_n ame	city	coun try	com pany _nam e	ener gy_m wh	water _m3	co2_t ons	stake hold er_n ame	stake hold er_ty pe	impa ct_ty pe	sever ity
1	2023	10	AMS- DC1	Amst erda m	NL	Goog le	5000 00	1200 00	4500 0	Resid ents	local_ resid ent	finan cial	high
1	2023	10	AMS- DC1	Amst erda m	NL	Goog le	5000 00	1200 00	4500 0	Resid ents	local_ resid ent	noise	medi um
1	2023	11	AMS- DC1	Amst erda m	NL	Goog le	5000 00	1200 00	4500 0	Muni cipalit y	muni cipalit y	regul atory	medi um
1	2024	10	AMS- DC1	Amst erda m	NL	Goog le	5200 00	1250 00	4300 0	Resid ents	local_ resid ent	finan cial	high
2	2023	10	AMS- DC2	Amst erda m	NL	Micro soft	3000 00	8000 0	2800 0	Resid ents	local_ resid ent	noise	high
3	2023	12	RTD- DC1	Rotte rdam	NL	Goog le	4000 00	1000 00	3800 0	Gree n NGO	envir onme	envir onme ntal	high

											ntal_ngo		
3	2023	13	RTD-DC1	Rotterdam	NL	Google	400000	100000	38000	Utility Rotterdam	utility	financial	medium

PK: (dc_id, year, stakeholder_id, impact_type)

Version 3 - 2NF Fix

What did we fix for 2NF?

- Several columns only depended on part of that key
 - ex) company_name only depended on dc_id, stakeholder_name and stakeholder_type only depended on stakeholder_id

How we fixed?

- The fix was to split the table into four: DataCenter, Environmental Impact, Stakeholder, and Stakeholder Impact, so that every column in each table depends on that table's full primary key. Especially Stakeholder impact table used as bridge table between Data center & Stakeholder tables for many to many relationship.

DataCenter

dc_id	dc_name	city	country	company_name	capacity_mw	status
1	AMS-DC1	Amsterdam	NL	Google	200	operational
2	AMS-DC2	Amsterdam	NL	Microsoft	150	operational
3	RTD-DC1	Rotterdam	NL	Google	180	operational

Environmental Impact

record_id	dc_id	year	energy_mwh	water_m3	co2_tons
1	1	2023	500000	120000	45000
2	1	2024	520000	125000	43000
3	2	2023	300000	80000	28000

4	3	2023	400000	100000	38000
---	---	------	--------	--------	-------

Stakeholder

stakeholder_id	stakeholder_name	stakeholder_type	city	country
10	Residents Amsterdam	local_resident	Amsterdam	NL
11	Municipality Amsterdam	municipality	Amsterdam	NL
12	Green NGO Rotterdam	environmental_ngo	Rotterdam	NL
13	Utility Rotterdam	utility	Rotterdam	NL

Stakeholder Impact

PK: (dc_id, stakeholder_id, impact_type)

dc_id	stakeholder_id	impact_type	severity	response
1	10	financial	high	protest
1	10	noise	medium	complaint
1	11	regulatory	medium	legal_challenge
2	10	noise	high	complaint
3	12	environmental	high	legal_challenge
3	13	financial	medium	complaint

Version 4 — 3NF Fix

What did we fix for 3NF?

- In the DataCenter table, company_name did not depend on dc_id directly, but depended on company itself. If we change the name of Google, we will have to change multiple rows.
- Also city and country were kind of duplicated in tables, so replaced to location

How we fixed?

- Extracted a Company table and a Location table, and replace the repeated columns with foreign keys pointing to those tables.

Company

company_id	company_name
1	Google
2	Microsoft

Location

location_id	city	country
1	Amsterdam	NL
2	Rotterdam	NL

DataCenter

dc_id	dc_name	capacity_mw	status	company_id	location_id
1	AMS-DC1	200	operational	1	1
2	AMS-DC2	150	operational	2	1
3	RTD-DC1	180	operational	1	2

Environmental Impact

record_id	dc_id	year	energy_mwh	water_m3	co2_tons
1	1	2023	500000	120000	45000
2	1	2024	520000	125000	43000
3	2	2023	300000	80000	28000
4	3	2023	400000	100000	38000

Stakeholder

stakeholder_id	stakeholder_name	stakeholder_type	location_id
7	Minseok	local_resident	1
8	Nikita	local_resident	1

9	Tasos	local_resident	1
10	Matvei	local_resident	2
11	Municipality Amsterdam	municipality	1
12	Green NGO Rotterdam	environmental_ngo	2
13	Utility Rotterdam	utility	2

StakeholderImpact

PK: (dc_id, stakeholder_id, impact_type)

dc_id	stakeholder_id	impact_type	severity	response
1	7	financial	high	protest
1	7	noise	medium	complaint
1	11	regulatory	medium	legal_challenge
2	9	noise	high	complaint
3	12	environmental	high	legal_challenge
3	13	financial	medium	complaint

Data Modelling

- Define Scope
 - We want to build a database to help politicians(or anyone interested) understand the impact of data center at each location to each stakeholder.
 - We want to know the answers for these questions by using this database
 - Where are data centers located?
 - What are the costs and waste?
 - What are the impacts on stakeholders?
- List how the data interacts in short sentences
 - Multiple Datacenters can be located in multiple locations(even in same location) so Location entities can have multiple data center name, but each data center can only have one location.
 - Each company can have 1 to many data centers

- One Data center can have multiple Environmental Impact(maybe environmental impact recorded yearly from each data center) 1 to many
- Location and stakeholders will have 1 to many relationship
- Identify the entities ("nouns", or core objects of your database)
 - Company
 - Data Center
 - Stakeholder
 - Stakeholder Impact
 - Environmental Impact
 - Location
- Define PK and FK
 - All entities need PK
 - location inside stakeholder, data center FK
 - company id inside data center FK
 - data center id inside environmental impact FK
 - data center id + stakeholder id + impact type = PK for stakeholder impact